1. DETERMINATION OF WEIGHT OF AN OBJECT USING THE PRINCIPLE OF MOMENTS

Aim:

To determine the weight of an object using the principle of moments

Apparatus required:

A metre scale, a knife edge, slotted weights, thread

Procedure:

i. A metre scale is supported at its centre of gravity by a knife edge or suspended by using a thread tied to its centre so that the scale is in the horizontal position. Ensure that the scale is in equilibrium position.

ii. A known weight W_2 and an unknown weight W_1 are suspended from to either side of the the scale using the weight hangers.

iii. Fix the position of one weight hanger and adjust the position of the second weight hanger such that the scale is in equilibrium.

iv. Measure the distance d_1 and d_2 of the two weight hangers from the centre of the scale accurately.

v. The experiment is repeated for different positions of the unknown weight. Measure the distances. The reading are tabulated as follows:

Observation:

S.No	Weight in the wieght hanger(W_2) kg	Distance of known wieght d_1 (m)	Distance of unknown wieght d_2 (m)	$W_2 \times d_2$ (kg m)	Unknown weight $W_1 = \frac{W_2 \times d_2}{d_1}$ (kg)
1					
2					
3					

Mean:

Calculations:

Moment of a force can be calculated using the formula

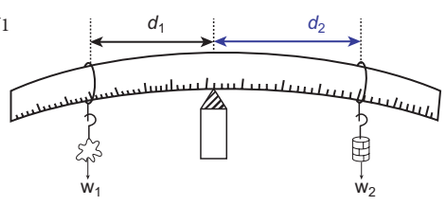
Moment of the force = Force x distance

Anticlock wise moment by unknown weight = $W_1 \times d_1$

Clockwise moment by known weight = $W_2 \times d_2$

$W_1 \times d_1 = W_2 \times d_2$

Unknown weight = $W_1 = \frac{W_2 \times d_2}{d_1}$



Result:

Using the principle of moments, the weight of the unknown body W_1 = ……. Kg Wt.

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